

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 683d4169 · Data hash 6d54...65b0

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 09:06 and 09:06 UTC. Reported results: final_val_loss 3.50806, avg_iteration_ms 125.668, tokens_per_second 32597. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	683d4169-a783-4e0b-9c0b-dc982c1feeac
Initialized	2026-10-07T09:06:15Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	2d079d310a3777c78784d14a6d98ea3b079170f2dcfb96c5205284e683ebb9e8
Disclosure	open
Source code, run 1	8faa6a451fb187f7fdd0bb3b9fd03f2858a9c6837b86e88616631dfc34080d62

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	14.9 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
val_loss	3.50806	74 (5 readings)	explicit
train_loss	3.39568	74 (74 readings)	explicit

Inline claim	Value	Phase	Output line
final_val_loss	3.50806		267
avg_iteration_ms	125.668		268
tokens_per_second	32597		269

4 Execution evidence

Coherence

Coherent (0.49): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	146	14.4 s	0.52 GB	99%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 51 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	09:06:16	f6c8042ce6903441	51
3	run_end	09:06:30	f6c8042ce6903441	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 56 files read, 0 files written, 2 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T09:06:31.269258+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	6d545071c552a8e5b2e6933585f65c1f436d8da78794d22c84622af96b5165b0
Code bundle hash	0b86756dddea1a345fcf61819d5fa17e41710704876aa7589f7462d266e23700

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
val_loss	4.50575	0	66	explicit
train_loss	4.28275	1	68	explicit
train_loss	4.83445	2	70	explicit
train_loss	4.40888	3	72	explicit
train_loss	3.60014	4	74	explicit
train_loss	3.65396	5	76	explicit
train_loss	3.91848	6	78	explicit
train_loss	4.03733	7	80	explicit
train_loss	3.56528	8	82	explicit
train_loss	3.89937	9	84	explicit
train_loss	3.50265	10	86	explicit
train_loss	3.93862	11	88	explicit
train_loss	3.70897	12	90	explicit
train_loss	3.93243	13	92	explicit
train_loss	3.78822	14	94	explicit
train_loss	3.63843	15	96	explicit
train_loss	3.42444	16	98	explicit
train_loss	3.93043	17	100	explicit
train_loss	3.73081	18	102	explicit
train_loss	3.48223	19	104	explicit
train_loss	3.59753	20	106	explicit
val_loss	3.66954	20	108	explicit
train_loss	3.81544	21	121	explicit
train_loss	3.70728	22	123	explicit
train_loss	3.21771	23	125	explicit
train_loss	3.42714	24	127	explicit
train_loss	3.80062	25	129	explicit
train_loss	3.59171	26	131	explicit
train_loss	3.8755	27	133	explicit
train_loss	3.54449	28	135	explicit
train_loss	3.40021	29	137	explicit
train_loss	3.29478	30	139	explicit
train_loss	3.19226	31	141	explicit
train_loss	3.46101	32	143	explicit
train_loss	3.60662	33	145	explicit
train_loss	3.49269	34	147	explicit
train_loss	3.71825	35	149	explicit
train_loss	3.39578	36	151	explicit
train_loss	3.24773	37	153	explicit
train_loss	3.1978	38	155	explicit
train_loss	3.63056	39	157	explicit
train_loss	3.09743	40	159	explicit
val_loss	3.60245	40	161	explicit
train_loss	3.90586	41	173	explicit

Metric	Value	Step	Line	Source
train_loss	3.39628	42	175	explicit
train_loss	3.37964	43	177	explicit
train_loss	3.45055	44	179	explicit
train_loss	3.62073	45	181	explicit
train_loss	3.1355	46	183	explicit
train_loss	3.45028	47	185	explicit
train_loss	3.96772	48	187	explicit
train_loss	3.25246	49	189	explicit
train_loss	3.44287	50	191	explicit
train_loss	3.77539	51	193	explicit
train_loss	3.75908	52	195	explicit
train_loss	3.43078	53	197	explicit
train_loss	3.4172	54	199	explicit
train_loss	3.29341	55	201	explicit
train_loss	3.27701	56	203	explicit
train_loss	3.0326	57	205	explicit
train_loss	3.49764	58	207	explicit
train_loss	3.31054	59	209	explicit
train_loss	3.53641	60	211	explicit
val_loss	3.51248	60	213	explicit
train_loss	3.21198	61	226	explicit
train_loss	3.45798	62	228	explicit
train_loss	3.40031	63	230	explicit
train_loss	3.43007	64	232	explicit
train_loss	3.62409	65	234	explicit
train_loss	3.04461	66	236	explicit
train_loss	3.30949	67	238	explicit
train_loss	3.68908	68	240	explicit
train_loss	3.31717	69	242	explicit
train_loss	3.66248	70	244	explicit
train_loss	3.61493	71	246	explicit
train_loss	3.75326	72	248	explicit
train_loss	3.84342	73	250	explicit
train_loss	3.39568	74	252	explicit
val_loss	3.50806	74	254	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	1	09:06:16	0.4 s	0.49 GB	58%, 4566 MB
summary	266	09:06:30	0.0 s	0.00 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	09:06:16	f6c8042ce6903441	51
1	phase train	09:06:16	f6c8042ce6903441	0
2	phase summary	09:06:30	f6c8042ce6903441	0
3	run_end	09:06:30	f6c8042ce6903441	0

Event	Path or command	Hash
read	kv_run.sh	
read	train_gpt2cu	
read	gpt2_124M_bf16.bin	
read	dev/data/tinyshakespeare/tiny_shakespeare_train.bin	
read	dev/data/tinyshakespeare/tiny_shakespeare_val.bin	
read	gpt2_tokenizer.bin	
read	.github/workflows/ci.yml	
read	.github/workflows/ci_gpu.yml	
read	.github/workflows/ci_tests.yml	
read	dev/cpu/matmul_forward.c	
read	dev/cuda/benchmark_on_modal.py	
read	dev/cuda/common.h	
read	dev/data/data_common.py	
read	dev/data/edu_fineweb.sh	
read	dev/data/fineweb.py	
read	dev/data/fineweb.sh	
read	dev/data/hellaswag.py	
read	dev/data/mmlu.py	
read	dev/data/tinyshakespeare.py	
read	dev/data/tinystories.py	
read	dev/download_starter_pack.sh	
read	dev/eval/export_hf.py	
read	dev/eval/run_eval.sh	
read	dev/eval/summarize_eval.py	
read	dev/loss_checker_ci.py	
read	dev/test/test_dataloader.c	
read	dev/test/test_outlier_detector.c	
read	dev/unistd.h	
read	dev/vislog.ipynb	
read	doc/layernorm/layernorm.c	
read	doc/layernorm/layernorm.py	
read	llmc/cublas_common.h	
read	llmc/cuda_common.h	
read	llmc/cudnn_att.cpp	
read	llmc/cudnn_att.h	
read	llmc/dataloader.h	
read	llmc/logger.h	
read	llmc/mfu.h	
read	llmc/outlier_detector.h	
read	llmc/rand.h	
read	llmc/sampler.h	
read	llmc/schedulers.h	
read	llmc/tokenizer.h	
read	llmc/utils.h	
read	profile_gpt2cu.py	
read	scripts/multi_node/run_gpt2_124M_mpi.sh	
read	scripts/pyrun_gpt2_124M.sh	
read	scripts/run_gpt2_124M.sh	
read	scripts/run_gpt2_1558M.sh	
read	scripts/run_gpt2_350M.sh	

Event	Path or command	Hash
read	scripts/run_gpt2_774M.sh	
read	scripts/run_gpt3_125M.sh	
read	test_gpt2.c	
read	train_gpt2.c	
read	train_gpt2.py	
read	train_llama3.py	
process	./train_gpt2cu	
process	awk { print; fflush() } /^step +[0-9]+v[0-9]+ \\\ loss/ { _match(\$0, /step +[0-...	

C Run history

Invocation 1, OK, 14.9 seconds, 269 output lines, started 2026-10-07T09:06:15

metric_hash 76fa18d2d2759e0fcef3d56ff527bdeff34f1fd2909c495b3808af9b9cca3427
run_digest c326e2a4f07f8119ef205264b919e3da0c62c6400e733c623db74041a166a7f4
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 6d545071c552a8e5b2e6933585f65c1f436d8da78794d22c84622af96b5165b0
signed_message_hash 28a67f43aad9dad6aabcbe66f0e081f6f1e7f453ab3067f9ef3b03354964b323
nonce 4115a87c15c563a6d25c059ef2a3b5c8
signed_at 2026-10-07T09:06:31.269258+00:00
sealed_at 2026-10-07T09:06:31.331575592Z
checkout_bundle 0b86756dddea1a345fcf61819d5fa17e41710704876aa7589f7462d266e23700

Signature (base64)

rsDW5A/qLwSw4ghKtFfteKFvIo8nddMv6KhYTIxYVfMBFg7FCnLdagNE1ZKj2YcBhKwc7Zof8AlviZ2TGGIZYzR+
C6RUGjMtjsoxVw4r0s4+mdniGfjwseJME/r7gUM3tEcLbtDBRMTc/8hG5BJiWau2LdII28jjoC0mqMR2XLYF00GK
dgKetWkiFhUtnUjuwy7uex8MB6pLEljKwcUqVt2J94IEVqdoaM4NHl9lFh3PIpx6YDYeIadhndaHj2BIyED/dm0F
mAhsw60aheFtYUTRA7phmjyJwV8Xvh5uDhINLSfQL74tqLWT1+MCNtgEaMrj4YGhfzhoL5zBrCX0Ec7qc6pzcxt
7Rtjw0h99dtqcATQep5sXDH4cLwfMODxxuXwmUKqfLZTdT2ajMV2QCeX2AWmzZ1JQNzvIopAEFdRKVELp//tzxdXG
RB6DAVyuCXXw90mSzYAk58+UNpr90I7gNq0SjiJ9YByQb1keA50QzLg4KQaPtMnc7qrJV33HedSQ2Dtri81fzvZ
tq2Vyf3YNdKFq486YmbA/Gpe83kHtdYJmfnYqtHIBosLI6yEM0NxfFeb066GUoTu6r/i7pAKlQNvJc7MLXsiezok/
RHlUQICicmc8KBd+aw3CNkWNpyqV6XH0IVi/0PbywzZC9y03qm4MyLOCJY91o00UqGY=

Public key

-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNnPpxYa5X
7qU95EFeisX/GfoS9ToXtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFcqDB5tNU7oHfYvBeNlrvPU9TAqdbtozdHQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKSvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syans67u7UMgIcNgeNHNcy9bLGPu5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JAPDXMNzNikaFfBsQK7iKxr7vIaTeZckFZwfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJrmB8LPP9v
8Jap/1PWjNcNku5cdw3gHLCaWEAAQ==
-----END PUBLIC KEY-----

Run 1 outputs

```
stdout 56de8a312d1a601fee8117f1826f615bb43a11c827a6a7f51d700b8f9cab406e
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 76fa18d2d2759e0fce3d56ff527bddef34f1fd2909c495b3808af9b9cca3427
digest c326e2a4f07f8119ef205264b919e3da0c62c6400e733c623db74041a166a7f4
```

kv_run.sh (unchanged)

```
pre 5202983f8a554f879f7bec2736f6d91a08ebbd79a9df5096cbd008a353344bab
```

```
post 5202983f8a554f879f7bec2736f6d91a08ebbd79a9df5096cbd008a353344bab
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```